### Illustrative one-pass model flow




The figure shows the single-pass inference path. The trained network produces a class and valid dimensions without testing candidate geometries at inference.

### Numerical limitations at the full boundary

The prediction is allowed to return exact 320 nm and zero-size geometries, as well as extremely narrow positive walls or gaps. For those cases, one forward HFSS verification may be too expensive to converge within the original 14-day data budget. Return the geometry and explicit unverified status rather than clipping it into a narrower domain.

### Categorical discontinuity

The same target can be realizable by disk, square or either ring, so the decision boundary between topology classes may be sharp. An 8-way classification head avoids averaging categorical labels, but a single continuous dimension head may still learn ambiguous regressions within the winning category. This remains a genuine limitation of the one-model, no-search constraint.

## 4. Fixed training protocol and objective

Training must be specified without trying different architectures, model families, loss weights or learning-rate grids. The following is a reproducible starting configuration, not a proven optimum.

| Parameter | Prespecified value |
|---|---|
| Optimizer | AdamW; learning rate 1 × 10^-3; weight decay 1 × 10^-4 |
| Epochs and batch size | 200 epochs; batch size 128 |
| Loss | Cross-entropy for 8 classes + Huber for normalized coordinates |
| Loss weights | 1.0 (classification) and 2.0 (dimension); fixed |
| Random seed | 42 for sampling, initialization and training |
| Training/held-out geometries | 64 / 16 geometric settings; all paired substrates stay together |
| Model selection | None: train once, no HPO, no architecture comparison |
| Evaluation | Held-out inverse-target error plus HFSS re-simulation of final predictions |

$$\mathcal{L} = \mathrm{CE}(\hat{t}, t^*) + 2\,\mathrm{Huber}(\hat{z}, z^*)$$

### Feasibility and non-uniqueness

Threshold requests can have several equally good geometric solutions or none at all. Domain endpoints introduce additional equivalent representations: solid = ring with zero inner dimension, and any zero-volume ring is a no-Au limit. Canonicalize physically equivalent labels before training. A fixed catalogue tie-breaker makes labels reproducible but not uniquely physical. Classification and dimension accuracy alone do not prove